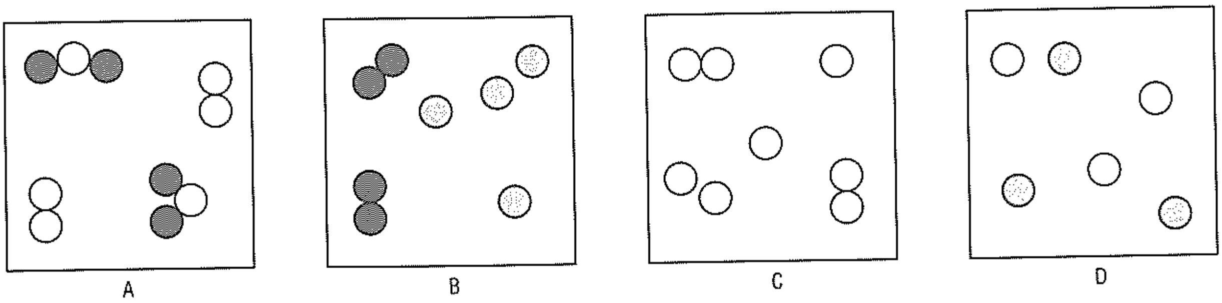
Atoms, molecules and ions—page 1

Name:

Skills: interpreting, understanding

Examine the four boxes below. Each circle represents an atom and the shading pattern represents different types of atoms.



Questions

1 Which of the boxes contains the following (note: it is possible that none of the boxes contain the requested item—if this is the case then write 'none'. It is also possible that more than one box may contain the requested item):

a individual atoms of elements only? ______

b chemical compounds only? ______

c chemical elements only? ______

d a mixture of compounds and elements? ______

2 When two atoms join together we can use the term 'diatomic molecule' to describe the molecule formed (from di = 2). When a chemical element exists as single atoms, the term 'monatomic' (from mon = 1) is used. Using the samples in boxes A to D above, draw examples of the following:

a two different types of diatomic elemental molecules

b two different monatomic elements

c a chemical compound

Atoms, molecules and ions—page 2

Name:

There has to be an EAZ ier way to represent atoms!

There is an easy way to represent atoms so that we can describe the number of protons, neutrons and electrons in any nucleus. This method is summarised by using the letters E to represent the element symbol, A to represent the mass number (number of protons and neutrons in the nucleus) and the atomic number (number of protons).

mass number → A

E ← symbol of the element

atomic number → Z

Use this method to complete the last column in the following table. A couple of examples have been done for you.

Atom	**Atomic number (number of protons)**	**Number of neutrons**	**Mass number**	${}^{A}_{Z}E$
Hydrogen	1	0	1	
Hydrogen-2 (deuterium)	1	1	2	${}^{2}_{1}H$
Carbon	6	6	12	
Oxygen	8	8	16	
Lead	82	126	208	${}^{208}_{82}Pb$
Uranium	92	146	238	

Extension

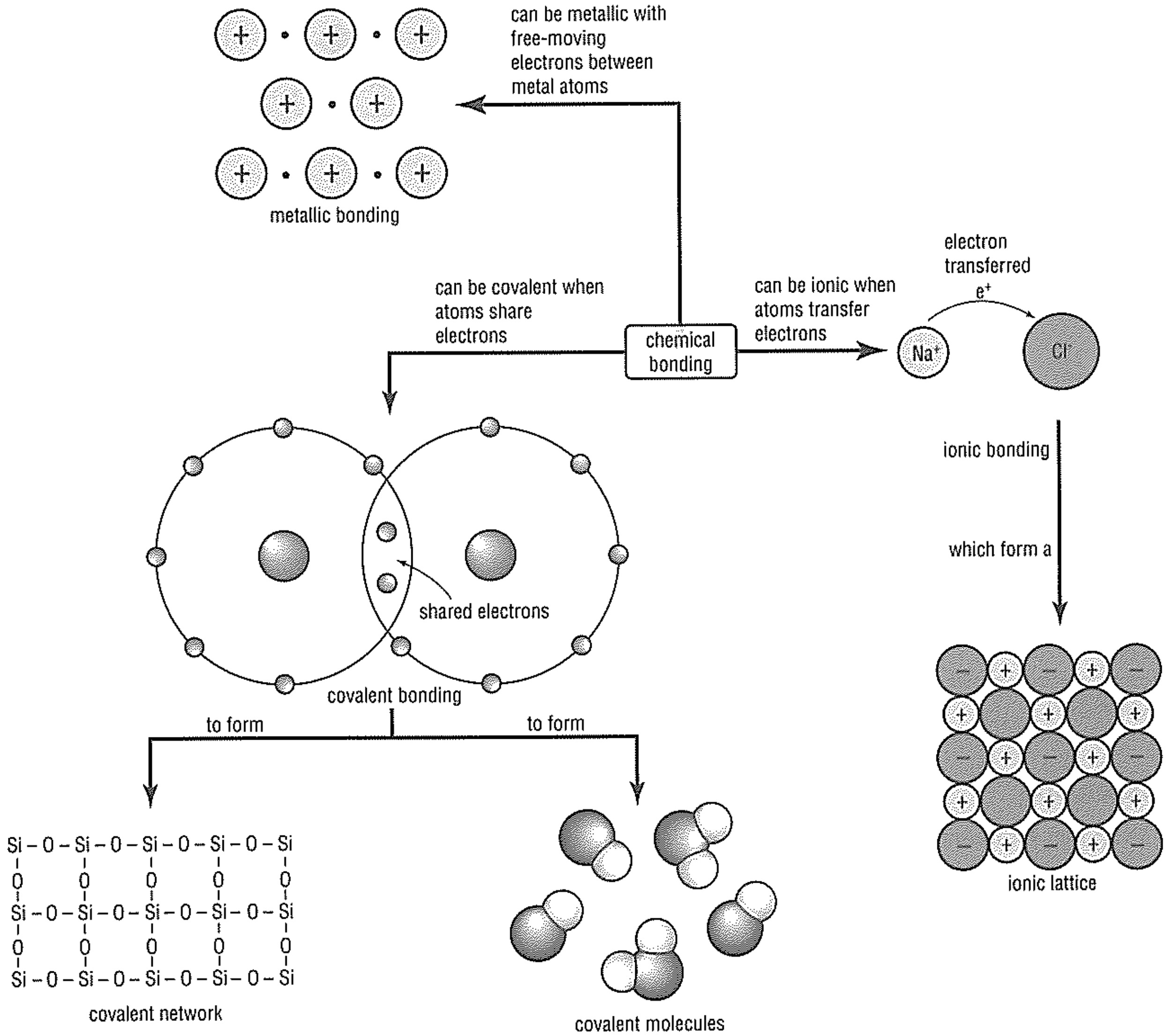
Given that for neutral atoms the number of electrons in the electron cloud is equal to the number of protons in the nucleus, add an extra column to the end of the table above to include the number of electrons you would find in a neutral atom in the examples above.

ACTIVITY 5.7 Chemical bonding—page 1

Skills: interpreting, knowledge

The concept map below summarises the bonding types explored in Focus 5.7. Review it carefully before you start. You may like to refer to Focus 5.7 when looking through the concept map.

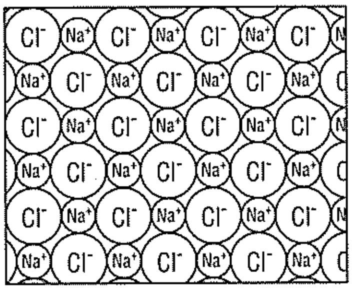
On the next page are some diagrams representing a range of substances. For each of the diagrams you are to identify the type of bonding as ionic, metallic, covalent network or covalent molecules. There is also some space under your answer to write a supporting statement for your decision. Discuss your answers with your teacher and class.



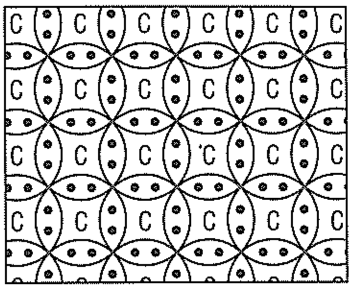
Chemical bonding—page 2

Name:

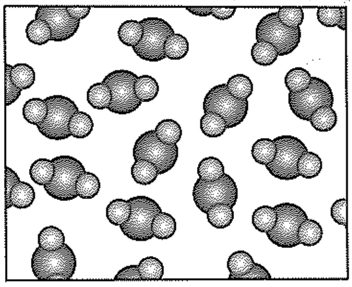
sodium chloride lattice



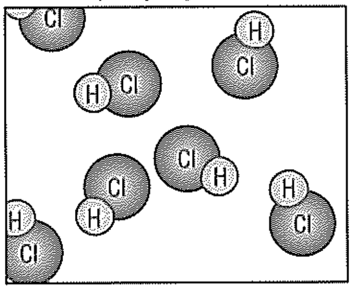
diamond lattice



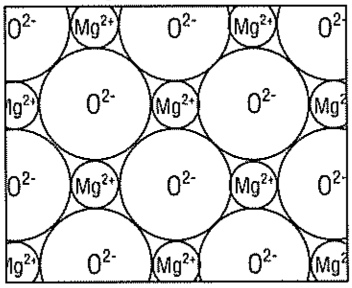
liquid water



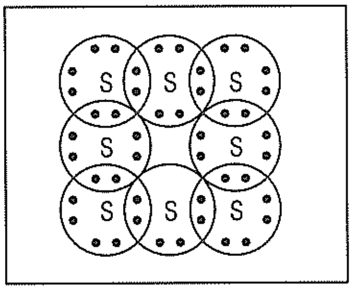
liquid hydrogen chloride



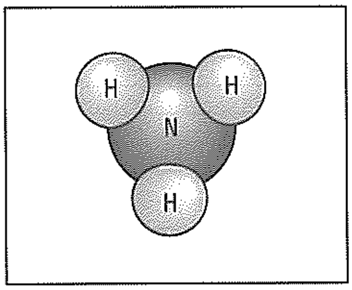
magnesium oxide lattice



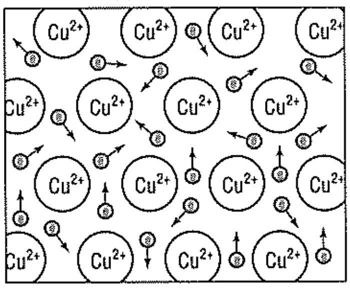
sulfur molecule



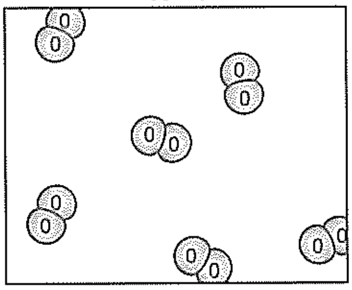
ammonia molecule



copper metal lattice



oxygen gas



The elements—page 1

Skill: knowledge

Atomic number	Symbol	Name	Atomic weight	Atomic radius*
1	H	hydrogen	1	37
2	He	helium	4	90
3	Li	lithium	7	125
4	Be	beryllium	9	90
5	B	boron	11	82
6	C	carbon	12	77
7	N	nitrogen	14	75
8	O	oxygen	16	73
9	F	fluorine	19	72
10	Ne	neon	20	71
11	Na	sodium	23	154
12	Mg	magnesium	24	136
13	Al	aluminum	27	118
14	Si	silicon	28	110
15	P	phosphorus	31	106
16	S	sulfur	32	102
17	Cl	chlorine	35	99
18	Ar	argon	40	95
19	K	potassium	39	203
20	Ca	calcium	40	174
21	Sc	scandium	45	144
22	Ti	titanium	48	132
23	V	vanadium	51	122
24	Cr	chromium	52	118
25	Mn	manganese	55	117
26	Fe	iron	56	117
27	Co	cobalt	59	116
28	Ni	nickel	59	115
29	Cu	copper	64	117
30	Zn	zinc	65	125
31	Ga	Gallium	70	126
32	Ge	germanium	73	122
33	As	arsenic	75	120
34	Se	selenium	79	117
35	Br	bromine	80	114
36	Kr	krypton	84	112
37	Rb	rubidium	85	216
38	Sr	strontium	88	191
39	Y	yttrium	89	162
40	Zr	zirconium	91	145
41	Nb	niobium	93	134
42	Mo	molybdenum	96	130
43	Tc	technetium	98	127
44	Ru	ruthenium	101	134
45	Rh	rhodium	103	125
46	Pd	palladium	106	128
47	Ag	silver	108	134
48	Cd	cadmium	112	148
49	In	indium	115	144
50	Sn	tin	119	141
51	Sb	antimony	122	140
52	Te	tellurium	128	136
53	I	iodine	127	133
54	Xe	xenon	131	131
55	Cs	cesium	133	235
56	Ba	barium	137	198
57	La	lanthanum	139	169
58	Ce	cerium	140	165
59	Pr	praseodymium	141	165
60	Nd	neodymium	144	164
61	Pm	promethium	145	163
62	Sm	samarium	150	162
63	Eu	europium	152	185
64	Gd	gadolinium	157	161
65	Tb	terbium	159	159
66	Dy	dysprosium	163	159
67	Ho	holmium	165	158
68	Er	erbium	167	157
69	Tm	thulium	169	156
70	Yb	ytterbium	173	144
71	Lu	lutetium	175	134
72	Hf	hafnium	178	136
73	Ta	tantalum	181	128
74	W	tungsten	184	126
75	Re	rhenium	186	127
76	Os	osmium	190	128
77	Ir	iridium	192	126
78	Pt	platinum	195	129
79	Au	gold	197	134
80	Hg	mercury	201	149
81	Tl	thallium	204	148
82	Pb	lead	207	146
83	Bi	bismuth	209	145
84	Po	polonium	209	
85	At	astatine	210	
86	Rn	radon	222	
87	Fr	francium	223	
88	Ra	radium	226	
89	Ac	actinium	227	
90	Th	thorium	232	
91	Pa	protactinium	231	
92	U	uranium	238	
93	Np	neptunium	237	
94	Pu	plutonium	244	
95	Am	americium	243	
96	Cm	curium	247	
97	Bk	berkelium	247	
98	Cf	californium	251	
99	Es	einsteinium	252	
100	Fm	fermium	257	
101	Md	mendelevium	258	
102	No	nobelium	259	
103	Lr	lawrencium	262	
104	Rf	rutherfordium	261	
105	Db	dubnium	262	
106	Sg	seaborgium	266	
107	Bh	bohrium	264	
108	Hs	hassium	277	
109	Mt	meitnerium	268	
110	Ds	darmstadtium	271	
111	Rg	roentgenium	272	
112	Uub	ununbium	285	

*Atomic radii are in picometres (pm).

The elements—page 2

Name:

Questions

Use the data on the previous page to answer these questions.

1 Give the symbols for the following elements:
- tungsten ______
- sodium ______
- zinc ______
- einsteinium ______
- tin ______
- boron ______
- cerium ______
- radon ______

2 Give the names for the following elements:
- Am ______
- Uub ______
- Ne ______
- Ar ______
- Cd ______
- O ______
- H ______
- Si ______
- S ______

3 Identify the following atoms from the table, giving their names and symbols:

a the atom with an atomic number of 88

______

b the atom with an atomic weight of 39

______

c the atom with an atomic radius of 133

______

d the atom with the symbol Hg

______

e the atom with an atomic number of 16

______

f the atom with the symbol He

______

g the atom with atomic weight 108

______

4 The atomic weight of an atom is its mass compared with that of a carbon atom. If an atom has an atomic weight less than carbon it is lighter than carbon. If an atom has an atomic weight larger than carbon it is heavier than carbon.

a State the atomic weight of a carbon atom.

______

b Identify two atoms that are lighter than carbon.

______

______

c Identify two atoms that are heavier than carbon.

______

______

d Describe what happens to the atomic weight as the atomic number increases.

______

______

e Calculate how many times heavier a magnesium atom is than carbon.

______

______

f Calculate how many times lighter a helium atom is than carbon.

______

______

5 The atomic radius gives an indication of the size of an atom. The table does not give the atomic radius for all atoms. Use the data shown to give the name and symbol for the:

a smallest atom ______

b largest atom ______

The periodic table—page 1

Skills: interpreting, knowledge

H — chemical symbol
hydrogen — name of element
1 — atomic number

	Group I	Group II											Group III	Group IV	Group V	Group VI	Group VII	Group VIII
	alkaline metals	alkaline earth metals	transition metals										non-metals				halogens	noble gases
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period 1	H hydrogen 1																	He helium 2
Period 2	Li lithium 3	Be beryllium 4											B boron 5	C carbon 6	N nitrogen 7	O oxygen 8	F fluorine 9	Ne neon 10
Period 3	Na sodium 11	Mg magnesium 12											Al aluminium 13	Si silicon 14	P phosphorus 15	S sulfur 16	Cl chlorine 17	Ar argon 18
Period 4	K potassium 19	Ca calcium 20	Sc scandium 21	Ti titanium 22	V vanadium 23	Cr chromium 24	Mn manganese 25	Fe iron 26	Co cobalt 27	Ni nickel 28	Cu copper 29	Zn zinc 30	Ga gallium 31	Ge germanium 32	As arsenic 33	Se selenium 34	Br bromine 35	Kr krypton 36
Period 5	Rb rubidium 37	Sr strontium 38	Y yttrium 39	Zr zirconium 40	Nb niobium 41	Mo molybdenum 42	Tc technetium 43	Ru ruthenium 44	Rh rhodium 45	Pd palladium 46	Ag silver 47	Cd cadmium 48	In indium 49	Sn tin 50	Sb antimony 51	Te tellurium 52	I iodine 53	Xe xenon 54
Period 6	Cs cesium 55	Ba barium 56	La* lanthanum 57	Hf hafnium 72	Ta tantalum 73	W tungsten 74	Re rhenium 75	Os osmium 76	Ir iridium 77	Pt platinum 78	Au gold 79	Hg mercury 80	Tl thallium 81	Pb lead 82	Bi bismuth 83	Po polonium 84	At astatine 85	Rn radon 86
Period 7	Fr francium 87	Ra radium 88	Ac actinium** 89	Rf rutherfordium 104	Db dubnium 105	Sg seaborgium 106	Bh bohrium 107	Hs hassium 108	Mt meitnerium 109	Ds darmstadtium 110	Rg roentgenium 111							

*Lanthanides 58–71	Ce cerium 58	Pr praseodymium 59	Nd neodymium 60	Pm promethium 61	Sm samarium 62	Eu europium 63	Gd gadolinium 64	Tb terbium 65	Dy dysprosium 66	Ho holmium 67	Er erbium 68	Tm thulium 69	Yb ytterbium 70	Lu lutetium 71
**Actinides 90–103	Th thorium 90	Pa protactinium 91	U uranium 92	Np neptunium 93	Pu plutonium 94	Am americium 95	Cm curium 96	Bk berkelium 97	Cf californium 98	Es einsteinium 99	Fm fermium 100	Md mendelevium 101	No nobelium 102	Lr lawrencium 103

The periodic table—page 2

Name:

Questions

1 On the periodic table, colour each of the following in different colours:
   a metals
   b non-metals
   c semi-metals (metalloids).

2 Identify which group consists of all gases.

3 What other name is given to the group VII elements?

4 State the atomic number for these elements:
   a platinum
   b gold
   c oxygen
   d magnesium
   e californium
   f bromine
   g krypton

5 State the name for atoms with the following atomic numbers:
   a 6
   b 3
   c 40
   d 13
   e 92
   f 56
   g 1

6 a Identify three elements that start with a C.
   b State the symbols for the elements you have chosen.
   c Explain why some elements have one letter for their symbol and others have two.

7 Outline a reason why elements are placed in the same group.

8 Explain what the atomic number tells us about an element.

9 Identify three elements that you think are named after famous people, and propose who that person was.

Ionic compounds, names and formulas—page 1

Skills: understanding, applying

Use the following table and examples to answer the questions.

Positive ions (cations)		Negative ions (anions)	
Formula	**Name**	**Formula**	**Name**
Mg^{2+}	magnesium	Cl^{-}	chloride
Na^{+}	sodium	CO_3^{2-}	carbonate
NH_4^{+}	ammonium	OH^{-}	hydroxide
Al^{3+}	aluminium	O^{2-}	oxide
Ca^{2+}	calcium	N^{3-}	nitride
Li^{+}	lithium	NO_3^{-}	nitrate
Cu^{2+}	copper(II)	SO_4^{2-}	sulfate
Cr^{3+}	chromium(III)	Br^{-}	bromide

Remember, the compound has to have an overall charge of zero, or it will not be stable.

Examples

Write a balanced compound formula for:

1 copper (II) chloride

Cu^{2+} Cl^{1-}

CuCl

$\boxed{CuCl_2}$

Start by writing the formulas.
Put the ion formulas together, without the charges. Always put the positive ion first.
Cross-multiply, putting the number of the charge on one ion, near the other ion, as shown.

Balanced compound formula

2 chromium (III) nitride

Following the above method, you end up with the formula Cr_3N_3. This is an incorrect formula. If the numbers next to each ion are the same, just use one of each.

$\boxed{CrN}$

3 ammonium carbonate

If more than one of a polyatomic ion is needed to balance the formula, use brackets.

$\boxed{(NH_4)_2CO_3}$

ACTIVITY 5.10

Focus 5.9: Chemical formulas

Ionic compounds, names and formulas—page 2

Name:

Questions

1 Name the following compounds:

a $MgCl_2$ ______

b $(NH_4)_2O$ ______

c Ca_3N_2 ______

d $CuCO_3$ ______

e $Cr_2(SO_4)_3$ ______

f $Cr(NO_3)_3$ ______

g $Al(OH)_3$ ______

2 Write formulas for these compounds:

a lithium carbonate ______

b aluminium bromide ______

c copper (II) nitrate ______

d sodium chloride ______

e calcium bromide ______

f ammonium hydroxide ______

g lithium nitride ______

3 Sodium chromate has the formula Na_2CrO_4.

a What is the formula of the chromate ion? ______

b What is the formula of:

i magnesium chromate? ______

ii aluminium chromate? ______

ACTIVITY 5.11

Writing formulas—page 1

Name:

Skills: knowledge, applying

Elements combine to form compounds. Each compound is represented by a unique chemical formula. A formula shows each element present, with subscripts stating how many atoms of each element are in the compound.

The formulas of molecules

1 Complete the following table with the names and formulas of common molecules that you should know. The atoms in these molecules are held together by covalent bonds.

Name	Formula	Name	Formula
nitrogen gas	N_2	hydrogen gas	
oxygen gas		water	
carbon dioxide		sulfur dioxide gas	
	CO		SO_3
	NO_2	chlorine gas	
nitrogen monoxide			F_2

The charges of ions

2 In order to write correct formulas for ionic compounds you need to know the charge on each ion involved. To help you learn these, complete the following tables of common ions.

Cations

+1		+2		+3	
	Na^+		Zn^{2+}	iron(III)	
hydrogen		copper		aluminium	
potassium		magnesium			
lithium			Ca^{2+}		
	Ag^+	iron(II)			

Anions

–1		–2		–3	
	Cl^-	oxide		nitride	
fluoride		sulfide		phosphate	
iodide			CO_3^{2-}		
bromide		sulfate			
nitrate					
	OH^-				
hydrogen carbonate (bicarbonate)					

ACTIVITY **5.11**

Writing formulas—page 2

Name:

Ionic formulas and names

Practise writing the formulas and names for ionic compounds by completing the tables below. If you need to revise how to do this, see Focus 5.9, or read the hint below.

Hint: When ions come together to form compounds, they combine in a ratio that gives the compound a total charge of zero. There must be enough negative charges to balance the positive charges and vice versa.

Sodium and chloride ions combine in a 1:1 ratio because sodium ions have a +1 charge and chloride ions have a −1 charge. Add these charges together: +1 + (−1) = 0. Thus, one of each ion join to give a compound with a total charge of zero. The formula is NaCl, and the name of this compound is sodium chloride.

Note that the charges of the ions are not included in compound formulas, but the numbers of each ion (the subscript numbers) are included. These subscript numbers indicate how many of each ion are in the formula. For instance, $MgCl_2$ indicates that there are one magnesium ion and two chloride ions in the formula. This is because magnesium (Mg) has a charge of +2 and chloride (Cl) has −1. Therefore +2 + (2 × −1) = 0. No charges appear in the overall formula because, once they are balanced, there is zero charge.

Using the cross method can help in writing formulas:

Place valency at the top of element	Cross over each valency number as subscripts	Remove valencies at top of element for final formula
$Al^{+3}\ O^{-2}$	$\overset{+3}{Al}_2 \leftrightarrows \overset{-2}{O}_3$	Al_2O_3

3

	Cl^-	F^-	NO_3^-	O^{2-}	CO_3^{2-}	PO_4^{3-}
Na^+					Na_2CO_3	
H^+	HCl					
Li^+						Li_3PO_4
NH_4^+				$(NH_4)_2O$		
Mg^{2+}						
Ca^{2+}				CaO		
Al^{3+}	$AlCl_3$					

4

Formula	Name	Formula	Name
$CaBr_2$		$FeCl_2$	
HCl		HNO_3	
$AlCl_3$		Ag_2SO_4	
Na_3PO_4		$NaHCO_3$	
CaO		CuS	
$(NH_4)_2CO_3$		$Fe(OH)_3$	

ACTIVITY 5.13

Revising chemical reactions—page 1

Name:

Skills: knowledge, interpreting

You should know the following reaction types. For more information about each type of reaction, see Focus 5.10.

- **Combination**: Two or more elements combine to form one compound.

$$A + B \rightarrow AB$$

- **Decomposition**: The opposite of a combination reaction—a complex molecule breaks down to make simpler ones. These reactions have the general form:

$$AB \rightarrow A + B$$

- **Neutralisation**: An acid and a base react with each other. Generally, the product of this reaction is a salt and water:

$$\text{acid} + \text{base} \rightarrow \text{salt} + \text{water}$$

- **Combustion**: Oxygen combines with a compound to form carbon dioxide and water. These reactions are exothermic, meaning they give out heat.

$$A + O_2 \rightarrow H_2O + CO_2$$

Questions

1 Using the above information, classify the following reactions according to their type.

Equation	Reaction type
$2Fe\ (s) + O_2\ (g) \rightarrow 2FeO\ (s)$	
$H_2O\ (l) \rightarrow H_2\ (g) + O_2\ (g)$	
$Na_2CO_3\ (s) \rightarrow Na_2O\ (s) + CO_2\ (g)$	
$NaOH\ (aq) + HNO_3\ (aq) \rightarrow NaNO_3\ (aq) + H_2O\ (l)$	
$HCl\ (aq) + NaOH\ (aq) \rightarrow NaCl\ (aq) + H_2O\ (l)$	
$CH_4\ (g) + 2O_2\ (g) \rightarrow CO_2\ (g) + 2H_2O\ (g)$	
$Pb\ (s) + O_2\ (g) \rightarrow PbO_2\ (s)$	
$NH_4OH\ (aq) + HCl\ (aq) \rightarrow H_2O\ (l) + NH_4Cl\ (aq)$	
$Ca(OH)_2\ (aq) + HCl\ (aq) \rightarrow CaCl_2\ (aq) + H_2O\ (l)$	
$C_{10}H_8\ (l) + 12O_2\ (g) \rightarrow 10CO_2\ (g) + 4H_2O\ (g)$	

2 Two of the above equations are not balanced. Identify the unbalanced equations, copy them into the space below and balance them.

Revising chemical reactions—page 2

Name:

3 For each of the following reactions:

i Identify the type of reaction taking place

ii Write a word equation for the reaction

iii Write a balanced chemical equation for the reaction

a Sulfuric acid reacts with potassium hydroxide to form water and a salt.

b Sulfur reacts with iron to form iron sulfide.

c Hydrogen peroxide (H_2O_2) breaks down to form oxygen gas and water.

d Barbecue gas (butane, C_4H_{10}) burns oxygen in an exothermic reaction.

ACTIVITY 5.14 Studying a reaction

Name:

Skills: numeracy, interpreting

Hydrochloric acid reacts with sodium carbonate, producing sodium chloride, water and carbon dioxide gas. The balanced chemical equation for this reaction is:

$$2HCl + Na_2CO_3 \rightarrow 2NaCl + H_2O + CO_2$$

A student conducted an experiment where 10 millilitres (mL) of acid (of constant concentration) was added to different masses of sodium carbonate. The student measured the volume of carbon dioxide gas produced in each case and recorded the results in a table:

Volume of HCl (mL)	Mass of Na_2CO_3 (g)	Volume of CO_2 produced (mL)
10.0	0.05	11.5
10.0	0.10	23.0
10.0	0.15	34.5
10.0	0.20	46.0
10.0	0.25	57.5
10.0	0.30	61.2
10.0	0.35	61.2
10.0	0.40	61.2
10.0	0.45	61.2

Questions

1 Plot a graph of the above data, showing the mass of sodium carbonate used on the horizontal axis, and the volume of carbon dioxide produced on the vertical axis.

2 Why do you think the volume of carbon dioxide produced did not increase after 0.3g of sodium carbonate was used?

Rates of reaction—page 1

Name:

Skills: knowledge, understanding

Part A

Use the list of words below to fill in the blanks. Some words may be used more than once, and some not at all.

concentration	products	catalyst	smaller
increase	reactants	kinetic	rate
reactants	collisions	ratio	higher
surface area	used up		

1 The ______________________ of reaction is a measure of how quickly ______________________ are turned into ______________________.

There are several ways to increase the rate of a reaction:

2 ______________________ the temperature. This will increase the ______________________ energy of the reactants, causing more collisions. These collisions will also have ______________________ energy and be more likely to cause a reaction.

3 Increase the ______________________ of a reactant. Having more reactant available causes more successful ______________________ between particles.

4 Increase the ______________________ ______________________ by breaking a solid reactant into ______________________ pieces. This allows more collisions between the reactants.

5 Add a ______________________ to speed up a reaction. A ______________________ helps a reaction proceed but is not ______________________ ______________________ in the reaction.

Part B

Monica and Ethan designed an experiment to test the effect of temperature on the rate of reaction using hydrochloric acid and magnesium. They used the following method:

Equipment: test tubes, water bath, magnesium ribbon, thermometer, stop watch, ruler

Method

Step 1 Add 20 ml of hydrochloric acid to a test tube.

Step 2 Warm the test tube of acid to exactly 20°C using a water bath at 20°C.

Step 3 When at the correct temperature add a 3 cm piece of magnesium ribbon to the test tube and start timing.

Step 4 Stop timing when the magnesium is completely dissolved.

Step 5 Repeat steps 1–4 at 40, 60, 80 and 100°C.

Rates of reaction—page 2

Name:

Results

Temperature (°C)	Time taken (seconds)
20	240
40	154
60	83
80	42
100	18

1 Plot these results on a graph, placing time on the horizontal (x) axis.

2 Describe any patterns and trends that you see in the results.

3 Use your graph to predict the time taken for magnesium to react at:

a 50°C

b 90°C

4 Use the results to draw a conclusion for the experiment.

5 Evaluate the experiment to decide whether it is a fair test.

6 Propose any improvements to the experimental method.

Writing and balancing chemical equations—page 1

Name:

Skills: interpreting, numeracy, knowledge

1 Balance the following equations:

a CH_4 (g) + O_2 (g) → CO_2 (g) + H_2O (l)

b H_3PO_4 (aq) + KOH (aq) → H_2O (l) + K_3PO_4 (aq)

c Al (s) + Cl_2 (g) → $AlCl_3$ (s)

d NH_3 (g) + O_2 (g) → NO (g) + H_2O (g)

e Mg + O_2 → MgO

f Mg + HCl → $MgCl_2$ + H_2

g Mg + N_2 → Mg_3N_2

h Fe + H_2SO_4 → $Fe_2(SO_4)_3$ + H_2

i $MgCO_3$ → MgO + CO_2

j C_2H_6 + O_2 → H_2O + CO_2

k $C_6H_{12}O_6$ + O_2 → H_2O + CO_2

2 Equations must be balanced so that the Law of Conservation of Mass holds true. State this law in your own words.

3 Use the information below to help answer the following questions.

Compound name	Compound formula	Compound name	Compound formula
Hydrochloric acid	HCl	Carbon dioxide	CO_2
Nitric acid	HNO_3	Calcium carbonate	$CaCO_3$
Sulfuric acid	H_2SO_4	Calcium nitrate	$Ca(NO_3)_2$
Magnesium chloride	$MgCl_2$	Magnesium hydroxide	$Mg(OH)_2$
Barium sulfate	$BaSO_4$	Barium nitrate	$Ba(NO_3)_2$
Sodium sulfate	Na_2SO_4	Sodium hydroxide	NaOH
Water	H_2O	Sodium carbonate	Na_2CO_3

Write balanced chemical equations for each of the following reactions.

Hint: follow these steps:

- Write the word equation for the reaction.
- Directly underneath the word equation, write the unbalanced formula equation.
- Balance the equation.

ACTIVITY 5.16

Writing and balancing chemical equations—page 2

Name:

a Hydrochloric acid is added to solid magnesium hydroxide, producing water and the soluble salt magnesium chloride.

b Nitric acid is added to solid calcium carbonate, producing bubbles of carbon dioxide, water, and the soluble salt calcium nitrate.

c Sodium hydroxide is added to sulfuric acid, producing water and the soluble salt sodium sulfate.

d Sulfuric acid is poured over solid sodium carbonate, producing carbon dioxide, water and the soluble salt sodium sulfate.

4 Write balanced equations for the following reactions. This time you will need to write formulas first.

a Iron metal reacts with chlorine gas to produce iron (III) chloride.

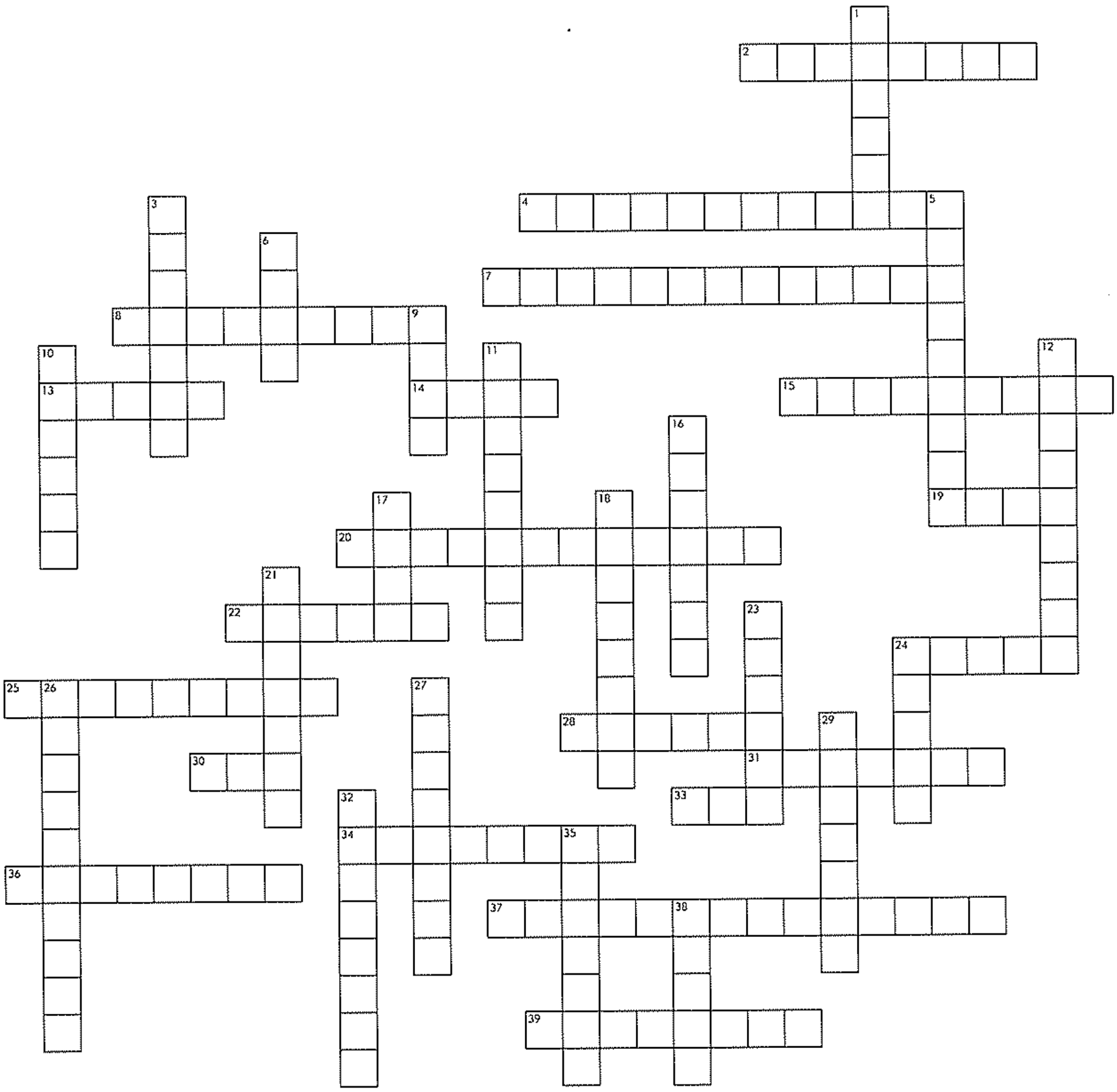
b Sulfur dioxide gas reacts with oxygen to produce sulfur trioxide gas.

ACTIVITY **5.17**

Natural and Processed Materials crossword—page 1

Name:

Skills: literacy, knowledge



Across

2 Colourless, odourless gas that is the lightest of all the chemical elements. It is also the most common element in the universe

4 Method of extraction that involves passing an electric current through a solution of the ore

7 Chemical opposite to composition reaction

8 Special substances that speed up the rate of a chemical reaction without themselves being used in the reaction

13 Type of bonding found in an ionic lattice

14 Smallest part that a chemical element can exist as

15 Can be hammered into thin sheets

19 Acid plus base produces a _______ plus water

20 If a large amount of acid is dissolved in water then the acid is said to be _____________

22 Can be polished to a shiny finish

24 Alloy of copper and zinc used to make a variety of house fittings

25 These are found on the left-hand side of a chemical equation